

# HELENA FU

(650) 863 - 4937 | [Helena.Fu@tufts.edu](mailto:Helena.Fu@tufts.edu) | [linkedin.com/in/helena-fu-ml](https://www.linkedin.com/in/helena-fu-ml) | [github.com/helena-f](https://github.com/helena-f)

## EDUCATION

### Tufts University

Medford, MA

B.S. Computer Science, B.A. Mathematics, Minor: AI Development

Jan. 2027

(GPA: 3.92 / 4.00)

Honors: 4x Tufts Dean's List, Al Gardner Memorial Endowment Scholarship (Society of Women Engineers)

Relevant Coursework: Algorithms, Data Structures, Operating Systems, Security, Machine Learning, Mathematical Modeling, Machine Architecture and Assembly

## TECHNICAL SKILLS

**Apple Platform:** SwiftUI, Swift, Xcode, Instruments, CoreBluetooth

**Performance & Systems:** CPU/memory profiling, flame graphs, WebSocket data streaming, real-time sampling, ETL pipelines

**Languages:** Swift, Python, JavaScript, TypeScript, Java, C++, SQL, HTML/CSS, Bash

**Frontend/Backend:** React, Next.js, Tailwind CSS, Node.js, Flask, Express, REST APIs, MongoDB

**DevOps/Cloud:** Docker, AWS, Google Cloud Platform, Palantir Foundry

**Developer Tools:** Git/GitHub, Jira, Asana, Tableau

## PROJECTS

### Instruments-Style Function Profiler | *Rust, React, D3.js, Python, Swift*

- Built an Instruments-style profiler (Rust + React/D3.js) with real-time CPU/memory sampling at 4Hz, interactive flame graph visualization, and multi-track timeline UI.
- Engineered a SwiftUI performance analyzer that detects 6 view inefficiency categories across SwiftUI/UIKit/CoreAnimation and surfaces severity-ranked diagnostics with Swift fix patterns.

### RobotPu Mobile Controller | *Swift, SwiftUI, CoreBluetooth, Xcode, Instruments, Gemini*

- Developed a SwiftUI iOS app to control a BBC micro:bit robot over Bluetooth with low-latency movement control.
- Profiled on-device LLM inference via Gemini using Instruments Allocations and Hangs, reducing command-to-response latency and achieving seamless real-time feedback.

### MIT Lincoln Lab Capstone – Autonomous Vehicle Awareness | *PyTorch, Python, Matlab*

- Led a team of 5 undergraduates to develop a 4D mmWave radar and camera fusion sensor system using CNN and GAN models for real-time scene reconstruction on a Raspberry Pi.
- Built a Next.JS and Flask dashboard to stream object detection, scene visualization, and performance profiling metrics.

## EXPERIENCE

### Mad Realities

New York City, NY

Software Engineering Intern

June 2025 – Aug. 2025

- Built LLM interface using Next.js, Gemini, Pinecone DB, and LangChain to streamline creative workflows for producers.
- Improved semantic search accuracy by 85% through embedding optimization, prompt engineering, and RAG fine-tuning.
- Built social media analytics dashboard with TypeScript, SQL, Palantir Foundry, reducing report generation time by 90%.

### Tufts Human-Robot Interaction Lab

Medford, MA

Research Assistant

Feb. 2024 – May 2025

- Extended robot architecture with a Java + Redis caching module, reducing natural language processing latency by 46%; profiled call paths before and after optimization to quantify hotspot elimination.
- Built a real-time preprocessing service in **Python**, improving cognitive model input quality by **30%**.

### MIT Schwarzman College of Computing

Cambridge, MA

Machine Learning Intern

Aug. 2024 – Dec. 2024

- Processed 11GB of unstructured marine data using NumPy, SQL, and ETL scripts, reducing model training time by 40%.
- Developed feature extraction and model stacking workflows using Scikit-learn, XGBoost and Random Forests to predict harmful algae bloom risk; improved baseline accuracy by 32%, reaching 76% overall.